ECON 0100 | Vignette C3 | Correctives

Solutions

Answers and work are shown in the red blocks.

After the report on the connection between Glittering Gum and Starlight Sneeze, the Ministry of Magic has tasked its economics branch to develop a policy to correct the problem.

Use a graph to propose a corrective policy and how it eliminates the DWL you found in Vignette C1.

Policy Type:	_____	Policy Size:	_____
Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Policy Quantity (Q^τ):	_____	Dead Weight Loss (DWL):	_____

Policy Type: a per-unit tax. Policy Size: 6 coins per piece of gum. Buyer's Price P_b: 18 coins. Seller's Price P_s: 12 coins. Post-Policy Quantity Q^τ: 24 pieces of gum. Dead Weight Loss: 0 (Vignette C1's 24 coins is eliminated).

From Vignette C1, demand is $P = 30 - \frac{1}{2}Q$ and supply is $P = 6 + \frac{1}{4}Q$. The market sets $30 - \frac{1}{2}Q = 6 + \frac{1}{4}Q$, so $Q = 32$ pieces of gum at $P = 14$ coins. Counting the Starlight Sneeze's 6 coins per piece (Vignette C1), $MSC = 12 + \frac{1}{4}Q$, and $30 - \frac{1}{2}Q = 12 + \frac{1}{4}Q$ gives the efficient $Q = 24$ at $P = 18$. The deadweight loss there was $\frac{1}{2} \times 6 \times (32 - 24) = 24$ coins.

The Starlight Sneeze is a negative externality, so the market makes too much gum. A tax equal to the external cost, 6, makes the buyer price sit 6 above the seller price:

$$30 - \tfrac{1}{2}Q = \left(6 + \tfrac{1}{4}Q\right) + 6 \quad \Rightarrow \quad Q = 24$$

Then $P_b = 30 - 12 = 18$ and $P_s = 6 + 6 = 12$. From the market's 14, buyers pay 4 more and sellers keep 2 less. The tax raises $6 \times 24 = 144$ coins.

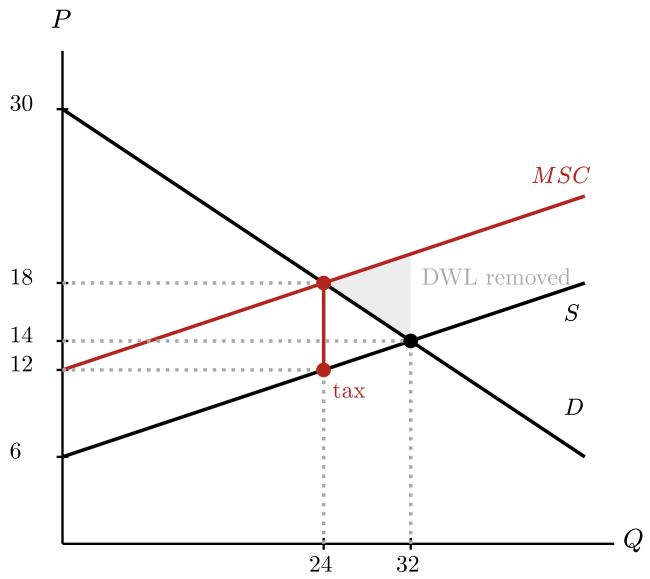


Figure 1: Corrective tax on the Glittering Gum curves

The market now stops at 24, the efficient quantity, so Vignette C1's deadweight loss of 24 is gone and the deadweight loss with the policy is 0: each piece of gum is made only if a buyer values it at least its cost to society, sneezing included. Buyers now face marginal social cost, because the tax adds the sneeze's 6 to every piece.